Tianruo Rose Xu

Mobile: +1 607-262-1199 Email: tx88@cornell.edu Website: xtrose29.github.io

Education

Cornell University, College of Engineering Aug 2024 – May 2027
B.S. in Computer Science Minor in Mathematics

Honors and Awards

- Selected for Cornell Computer Science nomination, CRA Outstanding Undergraduate Researcher Award, Fall 2026.
- Hunter R. Rawlings III Cornell Presidential Research Scholar.
- Clare Boothe Luce Undergraduate Research Award.
- Menschel Cornell Commitment Public Service Grant; Cornell Graduate School Opportunity Fund Grant.
- Bowers Undergraduate Research Experience (BURE) stipend and Academic Year Expansion support.
- USA Computing Olympiad (USACO) Platinum Division; full score in the January 2023 Gold Contest.
- USA Mathematical Olympiad (USAMO) Qualifier.

Publications and Manuscripts

[**Preprint**] **Tianruo Rose Xu**, Jiawei Ren, Yichi Yang, Zhaoxu Zheng, Lianhui Qin. *RT-SAFE: Benchmarking Agent Safety in Real-Time Embodied Environment.* arXiv preprint arXiv:2610.09294, October 2026. [Paper] [Project Website] [Code].

[**Preprint**] Haoqiang Kang, Yiming Zhang, Yiyang Guo, Chuying Li, Jianzhi Shen, **Tianruo Rose Xu**, Xiaokang Ye, Lianhui Qin. *DeliveryGym: An RL Environment for Long-Horizon Embodied Agent Planning with Adaptive Curriculum.* arXiv preprint arXiv:2609.19801, September 2026. [Paper] [Code].

[**Submitted**] *Can Emergent Verbalization of Epistemic Uncertainty Inform Factuality?*

[**Published**] **Tianruo Rose Xu**. *Can You Trust What I Think? Analyzing and Improving Verbalized Uncertainty and Factuality in Reasoning-Based Large Language Models.* AAAI-26 Undergraduate Consortium, Singapore, January 2026. [Poster] [Oral Presentation] [Paper].

[**Published**] **Tianruo Rose Xu**, Vedant Gaur, Liu Leqi, Tanya Goyal. *The Progress Illusion: Revisiting Meta-Evaluation Standards of LLM Evaluators.* Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China, November 2025. [Paper] [Poster].

[**Published**] Yingke Ding, Jiankai Tang, Wanying Mo, **Tianruo Rose Xu**, Yuanchun Shi, Yuntao Wang. *NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model.* MICCAI 2025 Workshop MSB EMERGE, July 2025. [Paper] [Code]. Runner-up Best Paper.

[**Published**] Tianruo Xu. *Elder Depression Detection by Multimodal Means.* BCP Education & Psychology, November 2022. [Paper].

Research Experience

Undergraduate Researcher, Cornell University

Computer Science Department, NLP Group; advised by Professor Tanya Goyal and Professor Claire Cardie.

Analyzing Verbalized Uncertainty and Factuality in Reasoning-Based LLMs 2025 – 2026

- Study whether epistemic uncertainty expressed in LLM reasoning can predict the factuality of individual claims in long-form responses.
- Led project design with Anirudh Atmakuru and Sanjana Ramprasad; built an end-to-end pipeline for reasoning-trace analysis, atomic fact extraction, trace-response alignment, FactScore evaluation, uncertainty analysis, and classifier training.
- Found that reasoning traces contain useful signals of factual reliability, suggesting that models' verbalized uncertainty can help identify potentially incorrect claims.
- Presented earlier work on reasoning-based uncertainty at the AAAI-26 Undergraduate Consortium; continuing the project in a manuscript currently under review.

Revisiting Meta-Evaluation Standards of LLM Evaluators 2025

- Investigated whether commonly used meta-evaluation methods reliably measure the quality of LLM-based evaluators.
- Led fine-grained experiments; built data and scoring pipelines, analyzed correlations with human

judgments, and tested sensitivity to reference models and evaluation choices.
- Found that apparent progress in LLM evaluators can depend strongly on the meta-evaluation setup, revealing limitations in commonly used evaluation standards.
- Contributed to experimental design, analysis, figures, interpretation, and manuscript writing for the EMNLP 2025 Findings paper.

Content and Style Evaluation in Natural Language Generation Systems **2024**
- Studied content quality and writing style as separate dimensions of natural language generation evaluation.
- Designed experiments and evaluation pipelines for model-based scoring, error analysis, and ablations across content and style criteria.
- Analyzed differences in evaluator behavior across the two dimensions to support more fine-grained evaluation of generated text.

Research Intern, UC San Diego

Computer Science Department, Q-Lab; advised by Professor Lianhui Qin.

RT-SAFE: Real-Time Embodied Agent Safety **2026**
- Proposed and led RT-SAFE, a benchmark for evaluating embodied-agent safety when the environment continues evolving during model inference rather than freezing the world.
- Defined the research direction, safety scenario taxonomy, benchmark tasks, real-time evaluation protocol, and metrics for passive and active safety failures.
- Led experiments across frontier vision-language models to study how inference latency and reasoning effort affect navigation safety.
- Found **12.3**× more collisions under real-time evaluation than static evaluation; although **94.1%** of episodes reached the goal, only **0.7%** finished without a safety event.
- Showed that longer reasoning can worsen safety because increased inference time leaves agents exposed to a changing environment for longer.
- Built the project website, released evaluation resources and code, and led analysis and writing for the preprint.

DeliveryGym / DeliveryBench: Embodied Agent Environments **2026**
- Developed traffic, obstacle, and safety infrastructure for long-horizon embodied-agent evaluation, including pedestrian traffic lights, signal-state checks, hazard evaluation, rollout support, and dynamic road conditions.
- Built and maintained the Unreal Engine-rendered first-person-view observation pipeline and visual assets used by embodied-agent rollouts.
- Extended the benchmark with the CityCore Paris environment by integrating semantic points of interest, buildings, road and navigation graphs, coordinate transformations, and 2D map assets.
- Developed traffic-signal and blocked-road variants to expand the range of dynamic urban scenarios available for evaluating planning and safety.

Undergraduate Researcher, Cornell University

Mathematics Department, Math + AI Lab; advised by Professor Daniel Halpern-Leistner.

Lean Proof Arena and LaTeX-to-Lean Autoformalization Pipeline **2026**
- Study how AI-generated mathematical proofs can be translated into Lean while preserving both formal correctness and the intended mathematical meaning.
- Develop a DARPA ExPMath-supported LaTeX-to-Lean autoformalization pipeline and benchmark tasks for research-level mathematical formalization.
- Build Lean Proof Arena for comparing multiple proof attempts and evaluating formal correctness, compilation success, and semantic fidelity.

Undergraduate Research Assistant, Tsinghua University

Department of Computer Science and Technology, Pervasive Human-Computer Interaction Lab.

Intelligent AI Tool Benchmarking and Orchestration Platform **2025 – 2026**
- Investigate how AI systems can match changing user goals to external tools using reusable capability representations rather than relying only on static tool descriptions.
- Develop a platform with standardized tool representations and multi-tool orchestration for academic

and general-purpose AI workflows.
- Designed and implemented back-end search and matching for tool retrieval, user intents, capability comparison, and personalized tool selection.
- Explore behavioral representations derived from tool interaction as a basis for more adaptive recommendation and selective probing of external tools.

NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model **2025**
- Supported development of a multi-task vision model combining segmentation, classification, and keypoint detection for automated nailfold capillaroscopy.
- Preprocessed data, trained baselines, ran experiments, evaluated model performance, and analyzed results across the component tasks.
- Contributed to the MICCAI 2025 Workshop MSB EMERGE paper, which received Runner-up Best Paper.

Undergraduate Research Assistant, Cornell University **2024**

Information Science Department; advised by Keigo Kusumegi and Professor Yian Yin.
- Processed large-scale Wikipedia data and applied K-means clustering to study how editor collaboration patterns relate to article quality.
- Analyzed collaboration clusters and article-level quality signals to characterize differences across editing patterns.

Teaching Experience

Cornell University
- **Head Teaching Assistant**, CS 3780: Introduction to Machine Learning Fall 2026
- **Climbing Instructor**, PE 1640: Basic Rock Climbing Fall 2026
- **Teaching Assistant**, CS 3780: Introduction to Machine Learning Spring 2026
  Designed the final Kaggle challenge, All About Birds!, a bird-sound spectrogram classification task with course-aligned baselines and evaluation.
- **Teaching Assistant**, CS 4820: Introduction to Analysis of Algorithms Fall 2025
- **Teaching Assistant**, CS 2800: Mathematical Foundations of Computing Spring 2025

Industry and External Research Experience

Incoming Machine Learning Engineer Intern, TikTok **2027**

TikTok E-commerce Knowledge Graph; NLP/CV/Multimodal/Algorithm. Seattle, WA

Independent Research Collaboration, Oracle **2025 – 2026**

Remote; mentored by Priyan Pattnayak, Senior Principal Scientist at Oracle Cloud.
- Received research mentorship on the AAAI work on uncertainty and factuality in reasoning-based language models.
- Currently investigate vision-language models for sign language understanding, including temporal continuity, spatial references, and robustness to interruptions in continuous signed interactions.